

Leet Code
top K elements
C Medium

1

Q Given a integer array nums and integer K, return the K most frequent elements. ~~You may~~

input = [1, 1, 1, 2, 2, 3], K = 2

Output = [1, 2]

Best Solution

Use a hashmap to count the number of

elements	Key	Value (Count)
	1	3
	2	2
	3	1

As hashmap can only store unique values and can be accessed using key, and sorted in $n \log n$ time

$O(n)$ - time

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Solution

class Solution

def topK Frequent elements(self, nums: list[int],
K: int) -> list[int]:

count = { } # Hashmap

for elements in nums:

count[element] = count.get(element, 0) + 1

Assigns Key and Value 0 + 1

sorted_count = sorted(count.items(),
key = lambda n: n[1],
reverse = True)

Sorts dict by value

top_k = []

for i in range(k)

key = sorted_count[i][0]

top_k.append(key)

Makes a list with keys which are sorted by value
return top_k

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First make a hash map for counting how many times a element appears in array.
(Key) (Value)

count = { }

Iterate through the list and assign the element as a key in count by count[element]

Then add the occurrence of the element by adding 1 everytime it appears by ~~count[element]~~

count.get(element, 0) + 1

for i in nums

count[element] = count.get(element, 0) + 1

Now sort the dictionary in descending order by value

sorted_count = sorted(count.items(), key = lambda
n: n[1],
reverse = True)

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Now make a list of top k elements by taking the first k ~~element~~ keys of ~~the~~ sorted dict in descending order

```
top_k = []
```

```
for i in range(k)
```

```
    key = sorted_count[i][0]
```

```
    top_k.append(key)
```

```
return top_k
```